

ASSIGNMENT 1

PART B – NOUN–VERB ANALYSIS

Course: B.Tech. CSE

Semester: 5

Subject: System Design (TCS-504)

System: Movie Ticket Booking System

Submission Date: 07-September-2026

B. NOUN–VERB ANALYSIS

The nouns from the problem statement are identified as possible classes. The verbs are used to identify the responsibilities and methods of those classes.

1. Nouns and Classes

Noun Found	Keep as a Class?	Reason
Movie	Yes	Has its own data and identity such as title, language and duration.
Seat	Yes	Has its own number, type and price.
Screen	Yes	Represents one auditorium and contains its physical seats.
Cinema	Yes	Represents the theatre and contains its screens.
Show	Yes	Represents a movie being shown on a screen at a particular time.
ShowSeat	Yes	Stores the status of one physical seat for one particular show.
Customer	Yes	Stores customer information such as name and phone number.
Booking	Yes	Represents a customer's booking with booking ID, seats, amount and status.
Payment	Yes	Represents the payment contract used to pay for a booking.

UPI Payment	Yes	Represents payment using UPI and implements the Payment contract.
Card Payment	Yes	Represents payment using a card and implements the Payment contract.
Cash Payment	Yes	Represents payment using cash and implements the Payment contract.
Price Calculator	Yes	Calculates the total price according to the selected seat types.
Ticket Printer	Yes	Formats and prints the ticket details.
Booking Service	Yes	Handles and coordinates the complete booking process.
Ticket	No	Ticket is the output printed from a confirmed Booking, so a separate class is not required.
Seat Layout	No	It is a view of the seats of a Show, not a separate entity. It can be displayed by a method.
Payment Method	No	The actual payment methods are represented by UpiPayment, CardPayment and CashPayment.

2. Verb → Methods / Responsibilities

Verb / Action	Class Responsible	Method
List / display movies	Cinema / MainMenu	<code>displayMovies()</code>
List shows	BookingService / Show	<code>showShows()</code>
Display seats	Show	<code>showSeatsLayout()</code>
Book seats	BookingService / ShowSeat	<code>bookTicket(), book()</code>
Check availability	ShowSeat	<code>isAvailable()</code>
Calculate price	PriceCalculator	<code>calculateTotal()</code>
Pay	Payment	<code>pay(amount)</code>
Print ticket	TicketPrinter	<code>printTicket()</code>
Cancel booking	BookingService / Booking	<code>cancelBooking(), cancel()</code>

3. Rejected Nouns

Some nouns are not kept as separate classes because they are better represented as data, output, or a responsibility of an existing class.

Ticket

Ticket is the output printed from a confirmed Booking. Therefore, a separate **Ticket** class is not required.

Seat Layout

Seat Layout is only a view of the seats associated with a Show. Therefore, it is handled through a display method rather than being represented as a separate class.

Payment Method

Payment Method is not kept as a separate class because the actual payment methods are represented by **UpiPayment**, **CardPayment**, and **CashPayment**.

4. Final Classes Identified

The final classes identified from the noun–verb analysis are:

1. Movie
 2. Seat
 3. Screen
 4. Cinema
 5. Show
 6. ShowSeat
 7. Customer
 8. Booking
 9. Payment
 10. UpiPayment
 11. CardPayment
 12. CashPayment
 13. PriceCalculator
 14. TicketPrinter
 15. BookingService
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